

Compensation of Amplitude Imbalance and Imperfect Quadrature in Resolver Signals for PMSM Drives

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Abstract—This paper proposes a compensation algorithm for reducing the position errors that are due to two nonideal output signals of a resolver in the vector control of a permanent magnet synchronous motor. Practically, a resolver generates periodic position errors because of the amplitude imbalance and imperfect quadrature between the two output signals of the resolver. As a result, the dq -axis currents of the synchronous reference frame have ripple components that are twice the stator fundamental frequency. In this paper, the effects of the position errors are analyzed on the basis of the synchronous dq -axis current equations, including the position errors. The d -axis current is directly used as the input signal of the proposed compensator to estimate the position errors by using a simple integral operation according to the rotor position, because the d -axis current is typically constant or zero. The proposed algorithm additionally considers the bandwidth effect of the closed current control loop to estimate the exact error signals according to the variation of the rotor speed. Therefore, the proposed algorithm does not need any additional hardware and much computation time. Furthermore, this algorithm can be applied to both the steady and transient states. The experimental results verify the effectiveness of the proposed algorithm.

Index Terms—Amplitude imbalance, bandwidth effect of the closed current control loop, imperfect quadrature, integral operation, permanent-magnet synchronous motor (PMSM), position errors, resolver.

I. INTRODUCTION

IN RECENT years, the use of the electric motors in automotive applications such as electric power steering and hybrid and electric vehicles has increased. Resolvers have been used for position sensors to provide the absolute position information in industrial applications for many years. Their robustness and reliability make them particularly suitable to harsh industrial environments [1]–[3]. The resolver output signals contain the

angular position information, and they have a digital form, which is obtained from a resolver-to-digital (R/D) converter. Several methods exist in the literature, focusing on ways to improve the measurement accuracy of the R/D converter [4]–[6]. These techniques are cost effective, and they can be implemented using a less hardware in order to reduce the weight and size. These methods state the position resolution and accuracy specifications under the assumption that the ideal resolver signals are supplied to an R/D converter. However, in a real system, the resolver output signals include the position errors in the resolver itself as well as in the resolver signal conditioning circuits [7]–[11]. As a result, the actual resolver output signals have nonideal characteristics such as amplitude imbalance, imperfect quadrature, inductive harmonics, reference phase shift, excitation signal distortion, and disturbance signals [7], [8]. Due to these nonideal characteristics of the resolver signals, the position information of the R/D converter can be considerably distorted. In particular, the amplitude imbalance and the imperfect quadrature are dominant components [7]–[11]. Therefore, this paper only focuses on the effects of the position errors caused by the amplitude imbalance and the imperfect quadrature. In order to solve these problems, the compensation algorithms have already been suggested [7]–[11]. In [7] and [8], these methods correct most of the nonideal characteristics, including an origin in the R/D converter. However, these methods require much labor and time, an excessive computation burden, and a hardware. In [9], this method introduces the gain-phase-offset-correction method only by a low computing effort. The method proposed in [10] was only presented to integrate the ideal rotor position to get the magnitude of the position error according to the distorted rotor position due to the amplitude imbalance, without considering the variation of the rotor speed and bandwidth of the closed current control loop. In [11], this method is just introduced to reduce the torque ripple caused by the amplitude imbalance. This method needs an additional position sensor which has no distorted position information in order to reduce the torque ripple. However, both effects of the amplitude imbalance and the imperfect quadrature must simultaneously be taken into account for the accurate vector control of the permanent magnet synchronous motor (PMSM) using a resolver.

In this paper, a new method is proposed for the compensation of the position errors caused by the amplitude imbalance and the imperfect quadrature between the resolver two output

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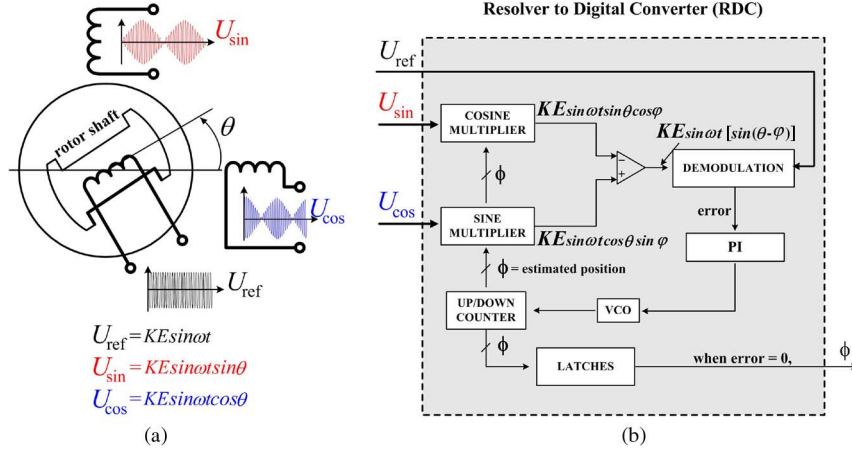


Fig. 1. Schematic diagrams of the (a) resolver and (b) tracking loop of an R/D converter with ideal resolver signals.

signals. The position errors are analyzed on the basis of the synchronous d - and q -axis current equations by using Park's transformation matrix and are composed of sine and cosine components that are twice the stator fundamental frequency. The d -axis current of the synchronous reference frame is used for the input signal of the proposed algorithm because the d -axis current is nearly constant or zero under all operating conditions. The phase delay between the d -axis current ripple and the rotor position is generated because it is directly related to the variation of the rotor speed and the bandwidth of the closed current control loop. Therefore, this paper presents the additional compensation method to reduce the position errors according to the modified rotor position, considering the bandwidth of the closed current control loop. The proposed algorithm is verified through several experiments.

II. ANALYSIS OF THE POSITION ERRORS

A. Ideal R/D Conversion [7], [8], [10]

The schematic diagrams of a resolver and an R/D converter are shown in Fig. 1. In Fig. 1(a), the resolver is basically a rotating transformer with one primary winding U_{ref} and two secondary windings U_{sin} and U_{cos} , respectively. The output signals of the resolver U_{sin} and U_{cos} are represented as follows:

$$U_{sin} = KE \sin \omega t \sin \theta \quad (1)$$

$$U_{cos} = KE \sin \omega t \cos \theta \quad (2)$$

where K is the transformer ratio, E is the excitation voltage applied on the primary winding, ω is the frequency of the excitation signal, and θ is the shaft angle to be measured.

Fig. 1(b) shows the tracking loop of an R/D converter, including the demodulation and the feedback loop to measure the rotor position. After the demodulation of the resolver output signals, the error voltage U_{err} can be derived by

$$U_{err} = KE \sin(\theta - \phi). \quad (3)$$

Finally, the error voltage U_{err} is forced to approach zero by recursively changing ϕ to track the real shaft angle θ in the tracking loop, as shown in Fig. 1(b). When this process is

finished, ϕ is outputted and used for the vector control of the PMSM as the digital position data.

B. Real R/D Conversion [7], [8]

Fig. 2 shows the typical measurement path of the resolver system. The resolver output signals are converted to digital values through a matching circuit, the low-pass filters, and an R/D converter, as shown in Fig. 2. The amplitude imbalance may be generated due to the unbalanced excitation and unequal inductance of the resolver signals and due to the nonideal characteristics of the analog devices. In addition, the imperfect quadrature between the two output signals of the resolver occurs when the two-phase windings are not machined or assembled with a perfect spatial quadrature. In Fig. 2, α and β describe the error values caused by the amplitude imbalance and imperfect quadrature, respectively.

Therefore, the resolver outputs with the amplitude imbalance and the imperfect quadrature errors can be expressed as

$$U_{sin} = KE \sin \omega t \sin(\theta + \beta) \quad (4)$$

$$U_{cos} = KE(1 + \alpha) \sin \omega t \cos \theta. \quad (5)$$

In (4) and (5), the error voltage after the demodulation in the R/D converter can be obtained as

$$U_{err} = KE [\sin(\theta - \phi) + \beta \cos \theta \cos \phi - \alpha \cos \theta \sin \phi]. \quad (6)$$

In (6), the tracking loop of the R/D converter drives U_{err} to zero. Therefore, (6) can be rewritten as

$$\sin(\theta - \phi) = -\beta \cos \theta \cos \phi + \alpha \cos \theta \sin \phi. \quad (7)$$

α , β , and the position error $\theta_{err} = \theta - \phi$ are small when the R/D converter appropriately operates. Using the aforementioned approximations, (7) can be used to derive

$$\theta_{err} = \frac{\alpha}{2} \sin(2\theta) - \frac{\beta}{2} (1 + \cos 2\theta). \quad (8)$$

As shown in (8), the position errors caused by the amplitude imbalance and the imperfect quadrature have ripple components that are twice the rotor position, and they are composed of sine and cosine terms.

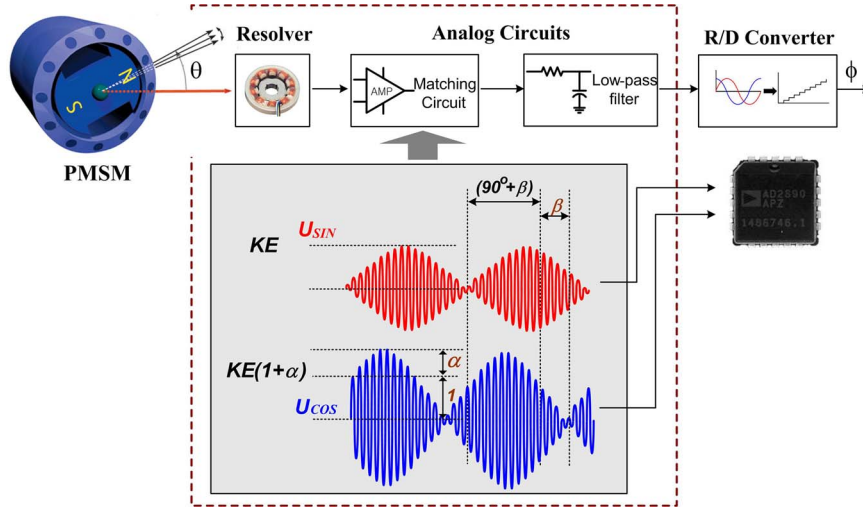


Fig. 2. Block diagram of the resolver position sensing system.

III. EFFECTS OF THE AMPLITUDE IMBALANCE AND IMPERFECT QUADRATURE IN PMSM DRIVES

A. Effect of the Position Error in PMSM Drives [10], [11]

The d - and q -axis currents of the stationary reference frame are given by

$$i_{ds}^s = i_{as} = -I_m \sin \omega_e t \quad i_{qs}^s = \frac{1}{\sqrt{3}}(i_{bs} - i_{cs}) = I_m \cos \omega_e t \quad (9)$$

where ω_e is the electrical angular velocity.

If there is a position error θ_{err} in the transformation matrix, the d - and q -axis currents of the synchronous reference frame can be derived by Park's transformation as

$$\begin{bmatrix} i_{ds}^e \\ i_{qs}^e \end{bmatrix} = \begin{bmatrix} \cos(\theta_e - \theta_{err}) & \sin(\theta_e - \theta_{err}) \\ -\sin(\theta_e - \theta_{err}) & \cos(\theta_e - \theta_{err}) \end{bmatrix} \begin{bmatrix} -I_m \sin \omega_e t \\ I_m \cos \omega_e t \end{bmatrix} \\ = \begin{bmatrix} -I_m \sin \theta_{err} \\ I_m \cos \theta_{err} \end{bmatrix} \quad (10)$$

where θ_e is the electrical rotor position.

As shown in (10), the position error causes the dq -axis current errors in the synchronous reference frame. If there is no position error in (10), the d -axis current must be zero.

B. Effects of the Position Errors Due to Both Amplitude Imbalance and Imperfect Quadrature

The transformation matrix, including both the amplitude imbalance and the imperfect quadrature errors (α and β), can be derived as shown by (11) at the bottom of the page. When β

is small, $\sin \beta$ and $\cos \beta$ can be approximated as $\sin \beta \approx \beta$ and $\cos \beta \approx 1$. As a result, the dq -axis currents i_{ds}^e and i_{qs}^e can be derived as

$$i_{ds}^e = I_m \left(-\frac{\alpha}{2} \sin 2\theta_e + \beta \frac{1 + \cos 2\theta_e}{2} \right) \quad (12)$$

$$i_{qs}^e = I_m + \alpha I_m \frac{1 + \cos 2\theta_e}{2} + \frac{\beta I_m}{2} \sin 2\theta_e. \quad (13)$$

If the d -axis currents of (10) and (12) have the same value, the error component of the d -axis current can be represented as

$$-\sin \theta_{err} = -\frac{\alpha}{2} \sin 2\theta_e + \beta \frac{1 + \cos 2\theta_e}{2}. \quad (14)$$

When θ_{err} is small in (14), the position error can be obtained as

$$\theta_{err} \approx \frac{\alpha}{2} \sin 2\theta_e - \beta \frac{1 + \cos 2\theta_e}{2}. \quad (15)$$

As known from (15), the first and second terms are the position errors caused by the amplitude imbalance α and imperfect quadrature β , respectively. As a result, the dq -axis currents in the synchronous reference frame have twice the stator fundamental frequency, as shown in (12) and (13).

C. Relationship Between the Resolver and the Motor Poles

In (8), the shaft angle θ that considers the number of resolver and motor poles can be rewritten as

$$\theta = (P_{resolver}/P) \times \theta_e = P_r \times \theta_e \quad (16)$$

$$\begin{bmatrix} i_{ds}^e \\ i_{qs}^e \end{bmatrix} = \begin{bmatrix} (1 + \alpha) \cos \theta_e & \sin(\theta_e + \beta) \\ -\sin(\theta_e + \beta) & (1 + \alpha) \cos \theta_e \end{bmatrix} \begin{bmatrix} -I_m \sin \omega_e t \\ I_m \cos \omega_e t \end{bmatrix} \\ = I_m \begin{bmatrix} \cos \theta_e (\sin \theta_e \cos \beta + \cos \theta_e \sin \beta) - (1 + \alpha) \sin \theta_e \cos \theta_e \\ (1 + \alpha) \cos^2 \theta_e + \sin^2 \theta_e \cos \beta + \sin \theta_e \cos \theta_e \sin \beta \end{bmatrix} \quad (11)$$

TABLE I
SUMMARY OF THE POSITION ERRORS IN EACH CASE ($p_{\text{resolver}}/p = p_r$)

$\begin{matrix} U_{\sin} \\ U_{\cos} \end{matrix}$	Normal	Amplitude imbalance	Imperfect quadrature	Both
Normal	0	$-\frac{\alpha}{2} \sin 2P_r \theta_e$	$-\frac{\beta}{2} (1 + \cos 2P_r \theta_e)$	$-\frac{\alpha}{2} \sin 2P_r \theta_e$ $-\frac{\beta}{2} (1 + \cos 2P_r \theta_e)$
Amplitude imbalance	$\frac{\alpha}{2} \sin 2P_r \theta_e$	0	$\frac{\alpha}{2} \sin 2P_r \theta_e$ $-\frac{\beta}{2} (1 + \cos 2P_r \theta_e)$	$-\frac{\beta}{2} (1 + \cos 2P_r \theta_e)$
Imperfect quadrature	$-\frac{\beta}{2} (1 - \cos 2P_r \theta_e)$	$-\frac{\alpha}{2} \sin 2P_r \theta_e$ $-\frac{\beta}{2} (1 - \cos 2P_r \theta_e)$	0	$-\frac{\alpha}{2} \sin 2P_r \theta_e$
Both	$\frac{\alpha}{2} \sin 2P_r \theta_e$ $-\frac{\beta}{2} (1 - \cos 2P_r \theta_e)$	$-\frac{\beta}{2} (1 - \cos 2P_r \theta_e)$	$\frac{\alpha}{2} \sin 2P_r \theta_e$	0

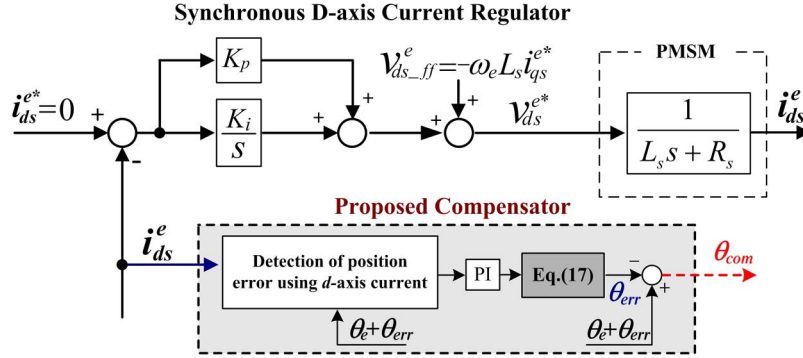


Fig. 3. Block diagram of the proposed compensator using the synchronous d -axis current.

where P_{resolver} is the number of resolver poles, P is the number of motor poles, and P_r is P_{resolver}/P .

In (8) and (16), the resultant position error, considering the number of resolver and motor poles, can be obtained by

$$\theta_{\text{err}} = \frac{\alpha}{2} \sin(2P_r \theta_e) - \frac{\beta}{2} (1 + \cos 2P_r \theta_e). \quad (17)$$

D. Torque Equation of the PMSM

The ideal torque equation of the PMSM is given by [10], [11]

$$T_e = \frac{3}{2} \frac{P}{2} \lambda_f i_{qs}^e. \quad (18)$$

The torque due to the position errors of the amplitude imbalance and the imperfect quadrature can be expressed as

$$T_e = \frac{3}{2} \frac{P}{2} \lambda_f I_m + \frac{3}{2} \frac{P}{2} \lambda_f I_m \left(\alpha \frac{1 + \cos 2P_r \theta_e}{2} \right) + \frac{3}{2} \frac{P}{2} \lambda_f I_m \left(\frac{\beta}{2} \sin 2P_r \theta_e \right). \quad (19)$$

Table I shows the summary of the position errors case by case. The column and row items provide four conditions of the resolver output signals $U_{\sin}$ and $U_{\cos}$, respectively.

IV. PROPOSED COMPENSATION ALGORITHM

Fig. 3 shows the block diagram of the proposed compensator using the d -axis current of the synchronous reference frame. In a typical vector-controlled PMSM drive, the d -axis current reference is usually set to zero. Therefore, the d -axis current is nearly constant or zero under all operating conditions. In Fig. 3, i_{ds}^{e*} is the d -axis reference current, v_{ds}^{e*} is the d -axis reference voltage, and $v_{ds_ff}^e$ is the feedforward d -axis voltage for the dq decoupling control. θ_{com} is the compensated rotor position produced by the proposed algorithm, as shown in Fig. 3. As known from (12), the d -axis current has the sine and cosine terms of twice the stator fundamental frequency. Therefore, the position error can be compensated by suppressing the ripple components of the d -axis current using the proportional-integral (PI) controller of the proposed compensator, as shown in Fig. 3.

A. Estimation Process Using the d -Axis Current

Figs. 4 and 5 show the integral operation of the d -axis current ripples according to the rotor position to estimate the position error. As shown in Figs. 4 and 5, α and β can be easily obtained by integrating the sine and cosine terms of (12), respectively. In Figs. 4 and 5, Integ-I and Integ-II are the integral results

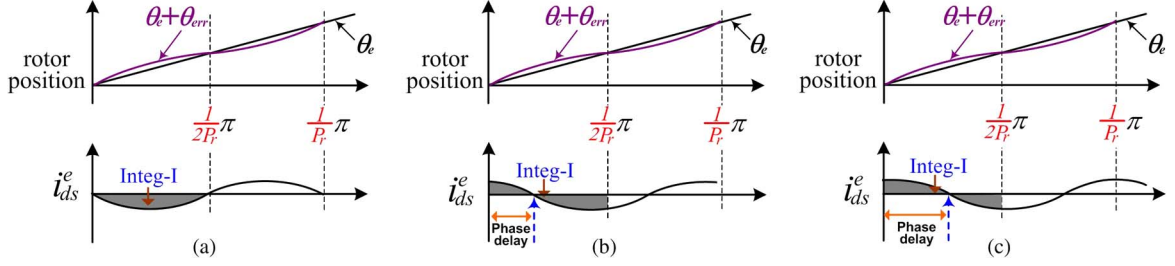


Fig. 4. In case of an amplitude imbalance, the integral operation process according to the rotor position. (a) Integrating the sine term. (b) Integ-I from the phase delay of the d -axis current at a medium speed. (c) Integ-I from the phase delay of the d -axis current at a high speed.

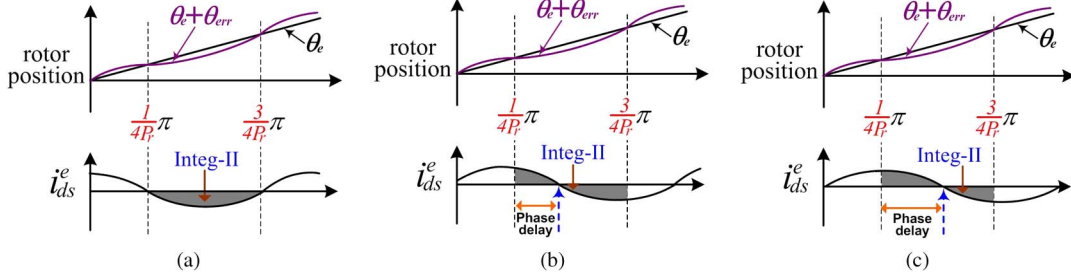


Fig. 5. In case of an imperfect quadrature, the integral operation process according to the rotor position. (a) Integrating the cosine term. (b) Integ-II from the phase delay of the d -axis current at a medium speed. (c) Integ-II from the phase delay of the d -axis current at a high speed.

of each integral section according to the rotor position. Integ-I can be acquired by integrating the d -axis current from 0 to $\pi/2P_r$. In the process, the cosine term is automatically removed by the integrating operation, as shown in Fig. 4(a). Integ-II can be simply obtained by integrating from $\pi/4P_r$ to $3\pi/4P_r$, as shown in Fig. 5(a).

B. Bandwidth of the Closed Current Control Loop According to the Variation of the Rotor Speed

In (12), the frequency of the d -axis current ripple depends on the rotor speed of the PMSM drive. The phase delay between the d -axis current ripple and the actual rotor position is caused by the bandwidth effect of the closed current control loop. As a result, the phase delay between the d -axis current ripple and the rotor position gradually increases according to the increase of the rotor speed, as shown in Figs. 4(b) and (c) and 5(b) and (c).

To correctly estimate α and β , the integral operation must be executed using an accurate rotor position. However, because of the phase delay of the d -axis current, the integration value of the d -axis current cannot guarantee an exact α and β . In order to obtain correct values of α and β , the modified rotor position θ_{modified} is required to get the constant position errors α and β according to the variation of the rotor speed.

The open-loop transfer function of the synchronous PI current regulator can be expressed as [12]

$$G_{OL}(s) = \left(K_p + \frac{K_i}{s} \right) \left(\frac{1}{L_s s + R_s} \right) \quad (20)$$

where K_p and K_i are the proportional and integral gains of the synchronous PI current regulator, respectively.

If K_p and K_i are selected as $K_p = L_s \omega_{cc}$ and $K_i = R_s \omega_{cc}$, the closed-loop transfer function of the d -axis current can be

approximated and thereby obtained as [13]

$$G_{CL}(s) = \frac{i_{ds}^e(s)}{i_{ds}^*(s)} = \frac{\omega_{cc}}{s + \omega_{cc}} \quad (21)$$

where ω_{cc} is the bandwidth of the closed current control loop.

Considering the bandwidth of the closed current control loop and the frequency of the d -axis current ripple, the phase delay can be calculated as

$$\begin{aligned} \Delta(\omega) &= -\tan^{-1} \left(\frac{\omega}{\omega_{cc}} \right) \\ &= -\tan^{-1} \left(\frac{2\pi \cdot f_{d\text{-current}}}{\omega_{cc}} \right) = -\tan^{-1} \left(\frac{2\pi \cdot nP}{60 \cdot \omega_{cc}} \right) \end{aligned} \quad (22)$$

where $f_{d\text{-current}}$ is the frequency of the d -axis current ripple and n is the rotor speed (in revolutions per minute).

As known from (22), the phase delay is proportional to the increase of the rotor speed and the decrease of the closed current control loop bandwidth. Therefore, the modified rotor position θ_{modified} is given in

$$\theta_{\text{modified}} = \theta_e - \tan^{-1} \left(\frac{2\pi \cdot nP}{60 \cdot \omega_{cc}} \right). \quad (23)$$

Therefore, the integral operation that is used to get α and β can be accurately obtained by considering the bandwidth effect of the closed current control loop.

C. Implementation of the Proposed Compensation Algorithm

In (12), the d -axis current has the sine and cosine terms and the ripple component that is twice the stator fundamental frequency. Therefore, the position error can be compensated by suppressing the ripple components of the d -axis current. Fig. 6 shows the overall block diagram of the PMSM drive with the

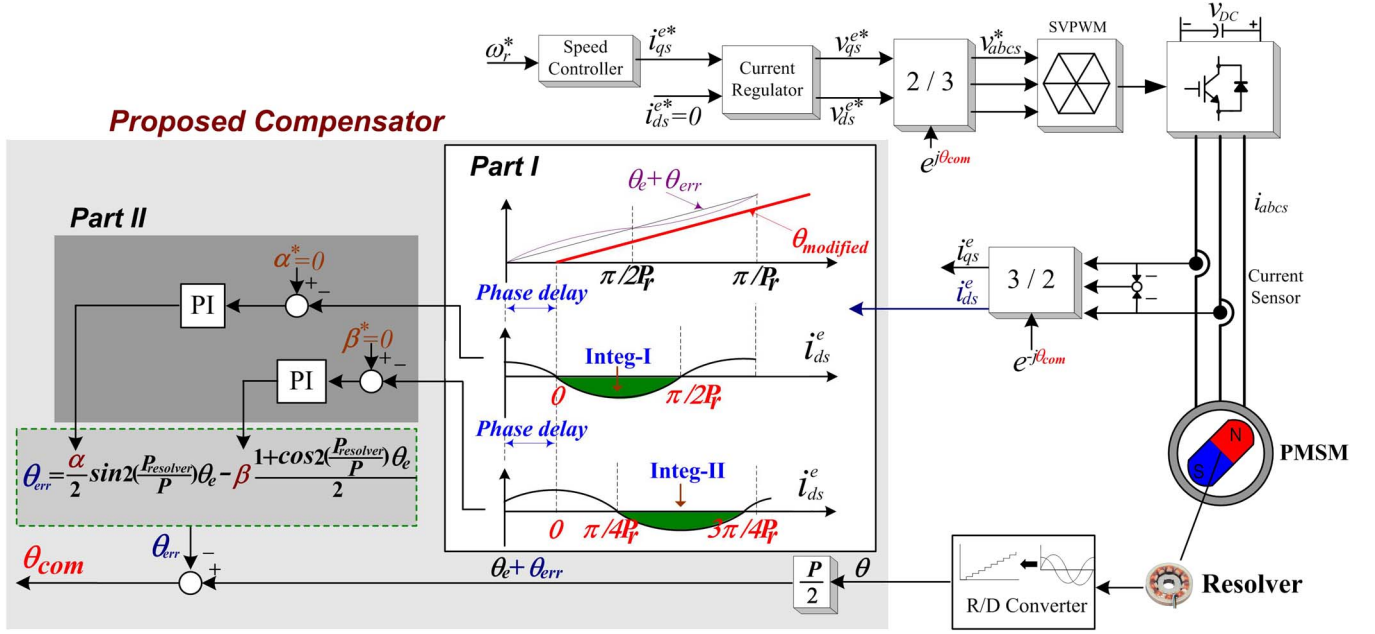


Fig. 6. Block diagram of the PMSM drive with the proposed compensator.

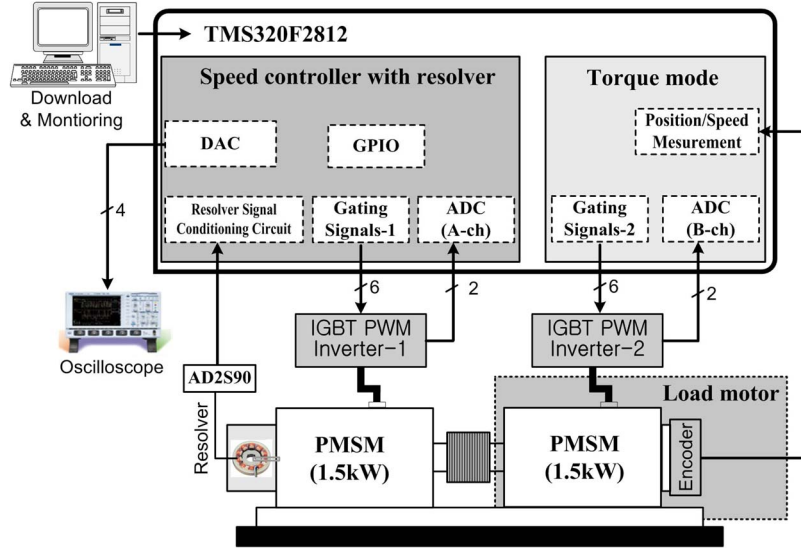


Fig. 7. Experimental PMSM drive system with the resolver and the load motor with an encoder.

proposed compensator. The proposed compensator consists of two parts, as shown in the rectangular box in Fig. 6. The first part is the integrator used for the estimation of the position errors from the synchronous d -axis current according to the modified rotor position θ_{modified} . The other part consists of two PI controllers to remove α and β . The PI gains of the proposed compensator affect the convergence rate of the position errors. In addition, the outputs of the proposed PI controllers are continuously updated to get the constants of the exact position error α and β . The compensated rotor position θ_{com} is achieved by subtracting the calculated position error θ_{err} from the R/D converter output. Therefore, the proposed algorithm does not need any hardware, and it is easily implemented with a software. Moreover, the structure of the proposed compensator is very simple, and a less computation time is provided, as shown in Fig. 6.

V. EXPERIMENTAL RESULTS

The proposed method was applied to the PMSM drive system and was implemented using a fixed-point DSP [14]. Fig. 7 shows the hardware configuration of the experimental PMSM drive system with the load motor. The other specifications of the PMSM and the resolver are shown in the Appendix. The current control period is set to 100 and 200 μs for the test of the bandwidth effect of the closed current control loop, respectively. For the transient state test, the load is generated from another PMSM that operates in the torque control mode, as shown in Fig. 7. To verify the proposed compensation algorithm, the amplitude imbalance and the imperfect quadrature errors were set to $\alpha = 5\%$ of the signal input range of the R/D converter and $\beta = 0.15$ rad/s, respectively. These values are approximately set from the experiments and references of the resolver and R/D converter used [15], [16].

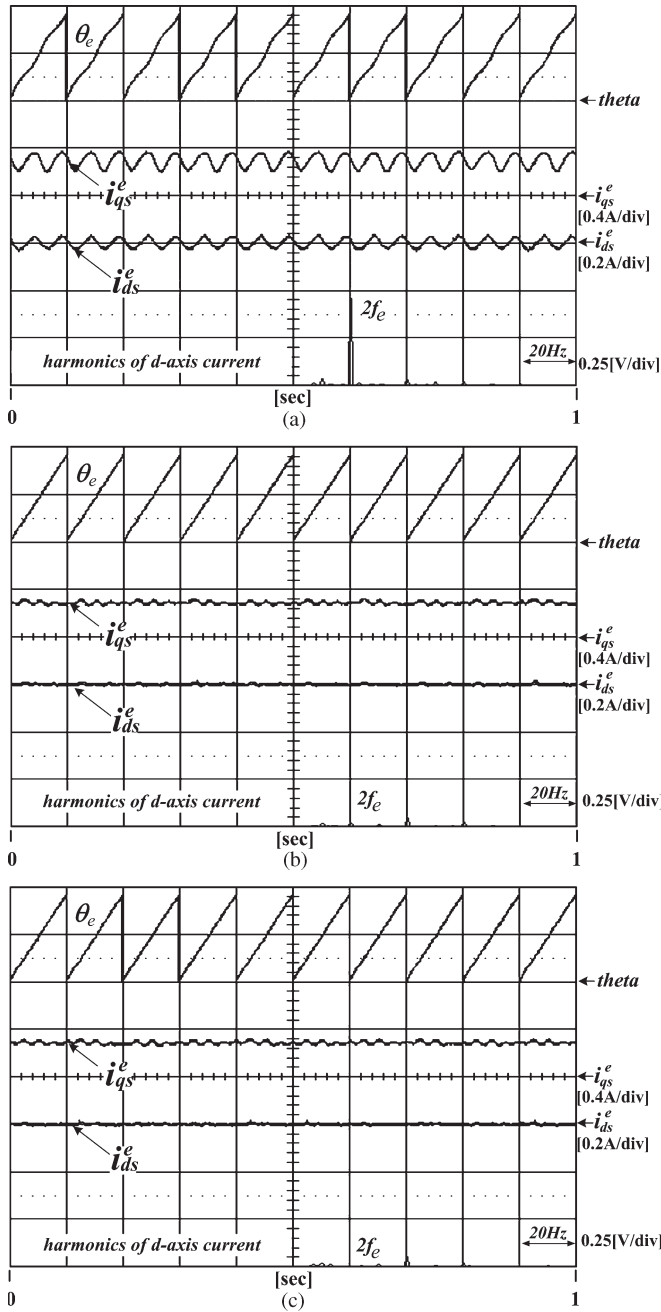


Fig. 8. Experimental results of the rotor position, current waveforms, DFT results (200 r/min), and sampling period of the current regulator (200 μ s). (a) Without the proposed algorithm. (b) With the proposed algorithm without considering the bandwidth of the closed current control loop. (c) With the proposed algorithm with consideration on the bandwidth of the closed current control loop.

Figs. 8 and 9 show the experimental results of the rotor position, the dq -axis currents, and the DFT results when the sampling period of the synchronous PI current regulator is 200 μ s and the motor is operated at 200 and 1500 r/min, respectively. Without the compensation algorithm, the rotor position is considerably distorted due to the position errors and the dq -axis currents that also have ripples that are two times the stator fundamental frequency, as shown in Fig. 8(a). At first, the dq -axis current ripples are almost decreased, without considering the bandwidth of the closed current control loop, as shown in Fig. 8(b). The twice ripple components of the d - and

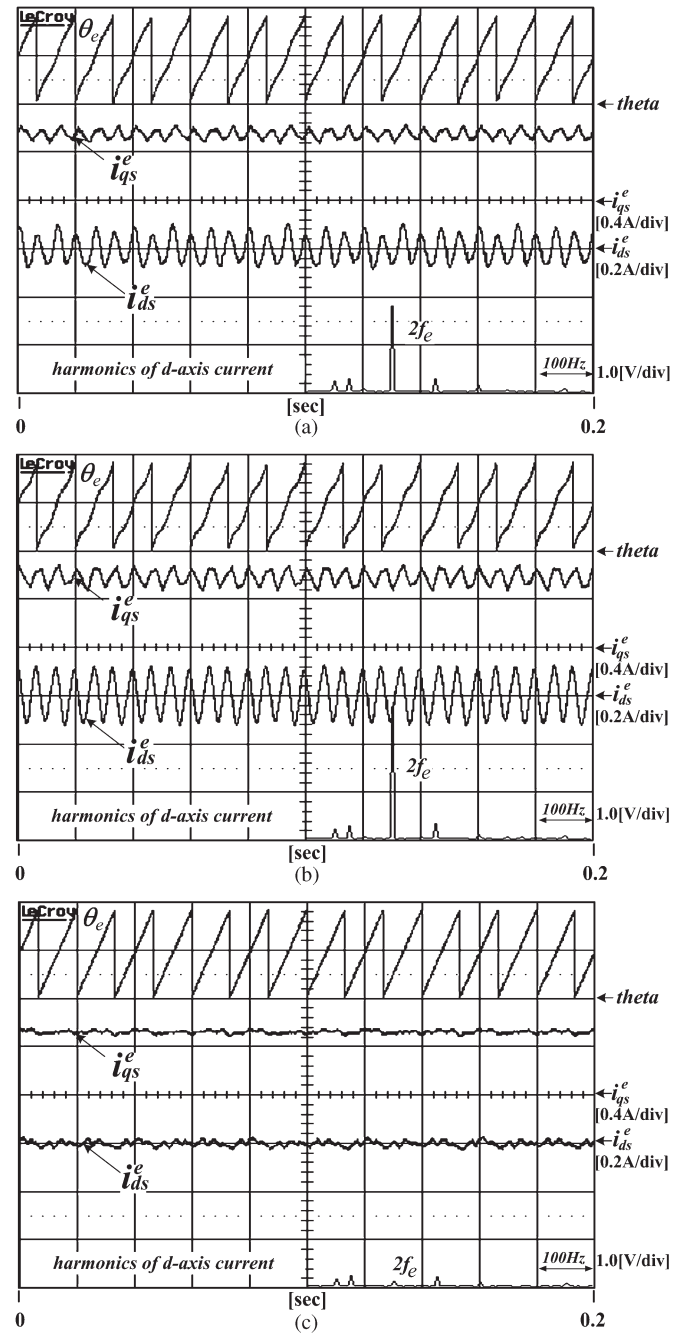


Fig. 9. Experimental results of the rotor position, current waveforms, DFT results (1500 r/min), and sampling period of the current regulator (200 μ s). (a) Without the proposed algorithm. (b) With the proposed algorithm without considering the bandwidth of the closed current control loop. (c) With the proposed algorithm with consideration on the bandwidth of the closed current control loop.

q -axis currents are also removed with the consideration of the closed current control loop bandwidth, as shown in Fig. 8(c).

Fig. 9 shows the experimental results at the same experimental conditions as shown in Fig. 8, except the rotor speed. As shown in Fig. 9(a), the rotor position and the d - and q -axis currents are seriously distorted due to the position errors.

Without considering the bandwidth effect of the closed current control loop, the dq -axis current ripples are inversely increased, as shown in Fig. 9(b). After considering the bandwidth effect of the closed current control loop, the distorted rotor

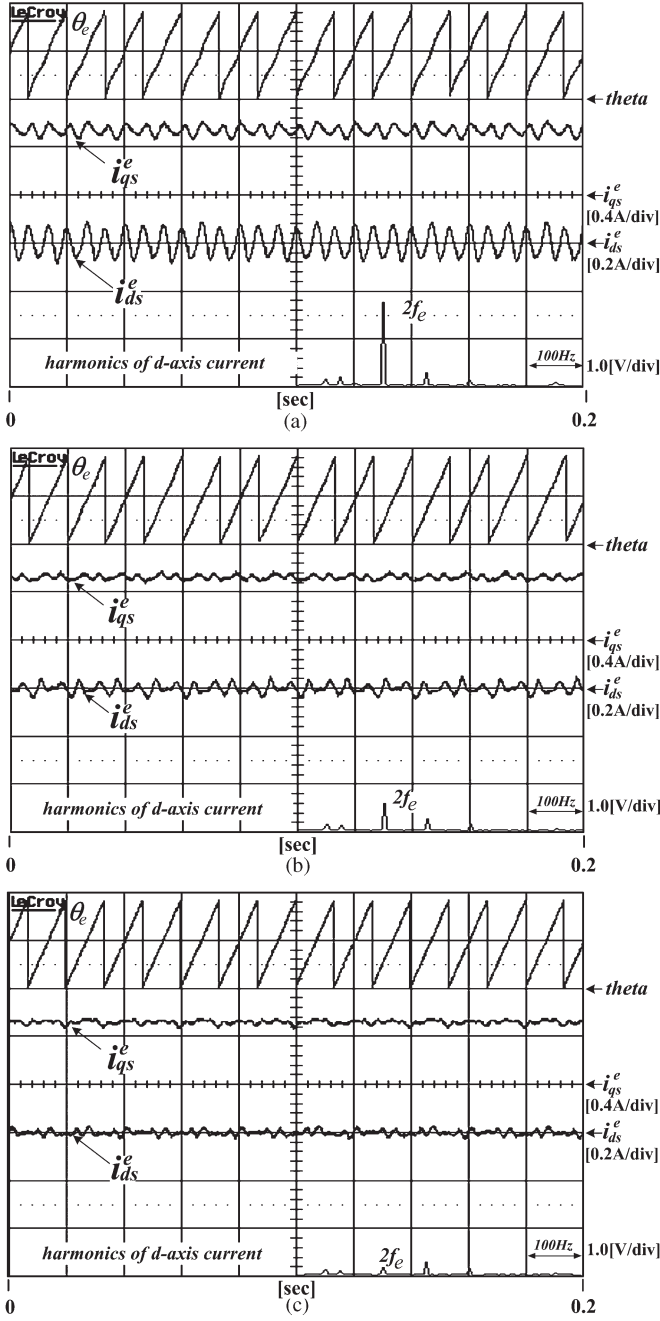


Fig. 10. Experimental results of the rotor position, current waveforms, DFT results (1500 r/min), and sampling period of the current regulator (100 μ s). (a) Without the proposed algorithm. (b) With the proposed algorithm without considering the bandwidth of the closed current control loop. (c) With the proposed algorithm with consideration on the bandwidth of the closed current control loop.

position and the dq -axis current ripples are nearly reduced to zero, as shown in Fig. 9(c), unlike the case in Fig. 8. As already mentioned in (22), the phase delay effect of the closed current control loop is very small in the low rotor speed range. However, the phase delay effect is significantly severe in the high speed range, as shown in Fig. 9.

The proposed compensation algorithm may be influenced by the bandwidth of the closed current control loop directly related with the sampling period of the current regulator. Thus, Fig. 10 shows the experimental results when the sampling period of the

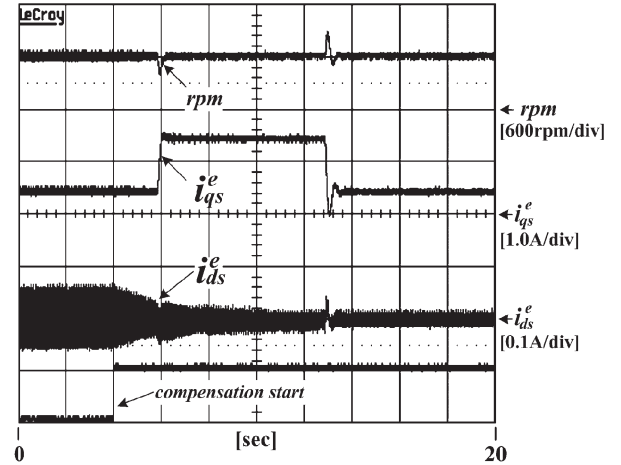


Fig. 11. Experimental results under the change of the load condition (0.5 N $\cdot$ m and 600 r/min) and sampling period of the current regulator (200 μ s).

PI current regulator is 100 μ s. As shown in Fig. 10(a), the rotor position and the d - and q -axis currents are still distorted by the position errors. Even though the bandwidth effect of the closed current control loop is not considered, the dq -axis current ripples are considerably decreased, as shown in Fig. 10(b), unlike the case in Fig. 9(b). In addition, considering the closed current control loop bandwidth, the distorted rotor position and the dq -axis current ripples are almost reduced to zero, as shown in Fig. 10(c).

In Figs. 8–10, the proposed algorithm can be effectively operated for the compensation of the position errors despite the change of the rotor speed and the bandwidth of the closed current control loop.

Fig. 11 shows the experimental results when the motor is operated at 600 r/min. The position errors can be compensated without a degraded control performance even in the transient state, as shown in Fig. 11. The given load torque is 10% of the rated torque and is generated by the same type of PMSM drive system used in the experiments.

Fig. 12 shows the experimental results of the speed waveform and the d - and q -axis currents according to the several steps of the rotor speed.

In Fig. 12(a), the amplitude of the d -axis current ripple is increasing because the rotor speed affects the ripple amplitude of the d -axis current. However, the amplitude of the d -axis current ripple is significantly decreased by the proposed compensation algorithm, as shown in Fig. 12(b). Therefore, the proposed method shows a good performance in a wide speed range, as shown in Figs. 8–12 by using the modified rotor position and by considering the bandwidth effect of the closed current control loop. The proposed compensation algorithm can be easily applied to both the steady and transient states, and it is verified through the several experimental results.

VI. CONCLUSION

A new compensation algorithm has been proposed to reduce the position errors caused by the amplitude imbalance and the

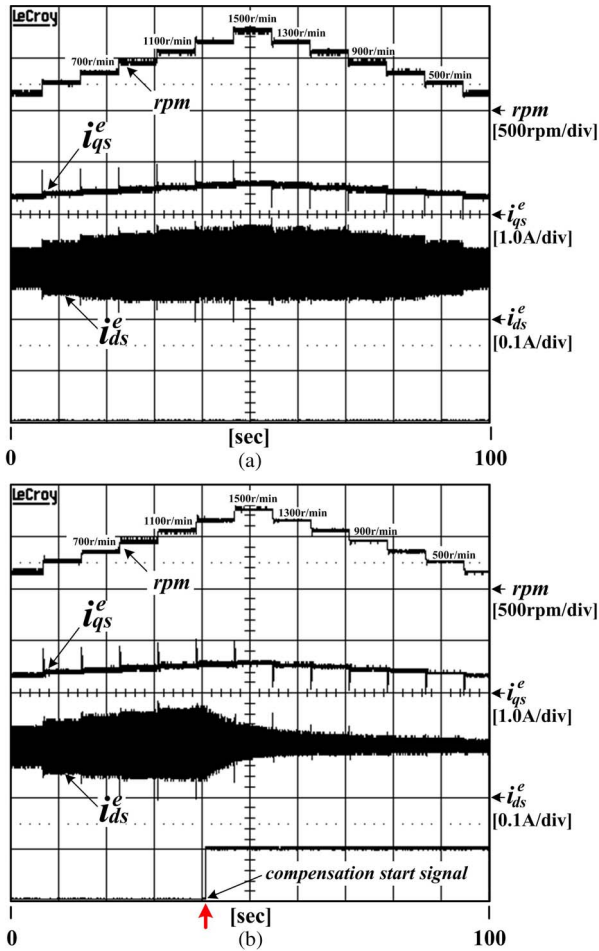


Fig. 12. Experimental results according to the change of the rotor speed from 300 to 1500 r/min and sampling period of the current regulator (200 μ s). (a) Without the proposed compensation algorithm. (b) With the proposed compensation algorithm.

imperfect quadrature between the two output signals of the resolver. The effects of the nonideal resolver output signals were analyzed on the basis of the synchronous d -axis current equation to derive the sine and cosine ripple components. The d -axis current was then used as the input signal of the proposed algorithm in order to estimate the position errors of the resolver output signals by using an integral operation according to the rotor position. In addition, the phase delay that is related with the rotor speed and the bandwidth of the closed current control loop was analyzed by using the transfer function equation of the synchronous PI current regulator. The proposed algorithm was implemented by using the d -axis current of the synchronous reference frame, the integrating operation to get the amplitude imbalance and the imperfect quadrature errors, and two PI controllers to remove the position errors. Therefore, the proposed algorithm provides no additional hardware and needs significantly less computational burden due to the simple integral operation. Moreover, the method can be applied to both steady and transient states because the d -axis current is usually constant or zero under all operating conditions.

The effectiveness of the proposed method was verified by several experimental results.

TABLE II
PMSM PARAMETERS

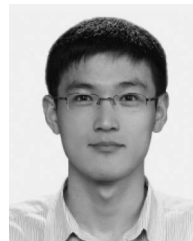
Rated capacity	1.5[kW]	Stator resistance	0.1024[Ω]
Rated current	7.5[A]	Stator inductance	2.027[mH]
Rated speed	3000[r/min]	back-EMF constant	0.017[V/rad/s]
Poles	6	Resolver poles	6

APPENDIX

See Table II.

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